Hui2Vec: Learning Transaction Embedding Through High Utility Itemsets

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Abstract. Mining frequent itemsets (FIs) in transaction databases is a very popular task in data mining. It helps create meaningful and effective representations for customer transactions which is a key step in the process of transaction classification and clustering. To improve the quality of these representations, previous studies have adapted vector embedding methods to learn transaction embeddings from items and FIs. However, FIs are still a simple pattern type that ignores important information about transactions such as the purchase quantities of items and their unit profits. To address this issue, we propose to learn transaction embeddings from items and high-utility itemsets (HUIs), a more general pattern type. Since HUIs were shown to be more appropriate than FIs for a wide range of applications, we take for hypothesis that transaction embeddings learned from HUIs will be more representative and meaningful. We introduce an unsupervised method, named Hui2Vec, to learn transaction embeddings by combining both singleton items and HUIs. We demonstrate the superior quality of the embedding achieved with the proposed method compared to the embeddings learned from items and FIs on four datasets.

Keywords: data mining · high-utility itemset mining · embedding methods · machine learning

1 Introduction

Transaction classification and clustering are very common data mining problems that have many applications in today's businesses mainly by supporting the decision-making process at different levels [2, 12]. When applied to the use case of a supermarket transaction dataset for example, these methods support the decision-making process with an enhanced and insightful customer profiling, a more accurate and informative market basket analysis, etc. These problems

have been well studied in the literature and many different methods have been proposed. Most of them require the use of machine learning techniques such as Support Vector Machines (SVM) and K-means.

Since using machine learning techniques usually requires input data of fixed size whereas there is no obligation for transactions to be of a fixed size, a transformation step is necessary here. That step is commonly referred to as feature extraction, a commonly used mechanism in machine learning. Feature extraction consists of transforming raw data into numerical features that can be processed while preserving the information of the original data set. Feature extraction usually yields better results than applying machine learning directly to the raw data.

To generate useful features for transaction data, a promising approach is to use frequent itemsets (FIs) as features [1, 2]. A frequent itemset is a set of values that frequently appears in transactions. After extracting FIs, each transaction can be converted into a vector of binary values, each indicating whether a specific frequent itemset is present in the transaction. Other more effective approaches introduced discriminative measures to handle the problem of high-dimensionality and data sparsity on datasets with a large number of FIs [3, 4]. Using discriminative measures helps extract only significant FIs but makes the feature extraction process supervised. These supervised FI-based approaches rely on the availability of labeled data. However, labels are not always provided and can be difficult to produce. The result is a representation tailored to a specific mining task that cannot be directly transferred to other tasks [5]

To cope with the limitations of FI-based and supervised FI-based approaches, Trans2Vec [5] was proposed as a new unsupervised embedding method for learning low-dimensional representations of transactions. Trans2Vec is inspired from embedding methods widely used in machine learning mainly in text and natural language processing [14–16]. It learns transaction embeddings from information of both singleton items and FIs. Trans2Vec was shown to outperform classical FI-based methods on several benchmark datasets [5].

The aforementioned FIs-based approaches are useful but an important limitation remains. It is that frequent itemset mining (FIM) algorithms for extracting FIs rely on a simplified transaction format where each item appears or not in each transaction. But in real-life, transactions data contain additional information such as the purchase quantities of items and their unit profits. To find itemsets that are profitable rather than frequent, FIM was generalized as high-Utility Itemset Mining (HUIM) [7–9]. The aim is to discover itemsets that have a high utility (importance) in transactions. The utility of an itemset can be measured according to several criteria such as the profit, frequency or weight. High utility itemsets (HUIs) and their variations were shown to be more appropriate than FIs for a wide range of applications such as market basket analysis, clickstream analysis [11], biomedicine [12], cross-marketing and mobile commerce [10]. Nonetheless, to our best knowledge, there are still no studies that have attempted to benefit from HUIs to create better transaction embeddings than using FIs.

In this paper, we present a new method to learn transaction embeddings through high utility itemsets, called Hui2Vec. It extends Trans2Vec to no longer operate at the level of FIs but at the level of HUIs. To extract HUIs, we adapt the EFficient high-utility Itemset Mining (EFIM) [6] algorithm, which was shown to be two to three times faster than many other state-of-the-art HUIM algorithms such as HUP-Miner [9] and FHM [8]. Th three main contributions of this study are:

1. We propose **Hui2Vec**, an unsupervised method, to learn low-dimensional continuous representations for transaction data.
2. We propose two models in **Hui2Vec**, which learn transaction embeddings from information of both singleton items and HUIs. We use EFIM [6], a very efficient state of the art algorithm to mine high utility itemsets.
3. We demonstrate that **Hui2Vec** is useful for transaction classification, as it achieves better results than Trans2Vec on several datasets.

2 Related Work

The proposed method, called Hui2Vec, is related to high-utility itemset mining (HUIM). The problem of HUIM has been widely studied in the literature. The traditional approach to HUIM is to discover high-utility itemsets in two phases using the Transaction-Weighted-Downward Closure model [20]. In the first phase, a set of candidate high-utility itemsets is generated by overestimating their utility. In the second phase, the database is scanned to to calculate the exact utilities of candidates and filter low-utility itemsets. Many algorithms follow this two-phase approach such as IHUP [7], Two-Phase [20], BAHUI [21], UP-Growth+ [10] and MU-Growth [22].

To overcome the problem of efficiency caused by the two very time-consuming phases, recent algorithms mine high-utility itemsets in a single phase. The first algorithms of this type are HUI-Miner and d^2HUP [19]. They avoid the problem of candidate generation and were reported to be 10 to 100 times faster than legacy two-phase approaches [19]. HUP-Miner [9] and FHM [8], two improved versions of HUI-Miner were proposed and shown to be 6 times faster than HUI-Miner. Despite all these improvements, high utility itemset mining approaches remain very expensive in time and memory consumption [8, 9, 19].

To go one step further into improving the efficiency of HUIM methods, EFIM, for Efficient high-utility Itemset Mining, a new one-phase algorithm was proposed [6]. The new approach introduced several new ideas that helped improve the performance both in memory consumption and execution time. EFIM design is based on the principle that all operations on each itemset should be performed in linear time. First, it introduced two efficient techniques to reduce the cost of database scans named High-utility Database Projection (HDP) and High-utility Transaction Merging (HTM). Second, it introduced two new upper-bounds on the utility of itemsets named the revised sub-tree utility and local utility to more effectively prune the search space. Finally, a novel array-based utility counting

technique named Fast Utility Counting (FAC) was used to calculate these upper-bounds in linear time and space for all extensions of an itemset. The authors conducted extensive experiments to compare EFIM to other state-of-the-art algorithms [6]. The results showed that EFIM is always performing better than other algorithms both in execution time and in memory consumption [6]. This is why we choose EFIM to extract high utility itemsets in the preparation step in Hui2Vec before we start learning the embedding.

The novel Hui2Vec approach is also related to vector embedding methods. These are learned representations of the data in the form of numerical vectors. The performance of machine learning methods crucially depends on the quality of these vector representations [23]. A wide range of methods for generating vector embeddings have been studied [23]. Among them, Word2Vec [14] is a well-known adaptation to learn embedding vectors for words in text.

Word2Vec inspired approaches to learn documents in text [15, 16] and approaches to learn node embeddings in graphs [27, 28, 24]. DeepWalk [27] and node2vec [28] are based on interpreting a sequence of nodes seen on random walks within a graph as if they were words appearing together in a sentence. Recent approaches are extending vector embeddings to learn graph representations [23, 26, 29, 30]. Trans2Vec [5] is an adaptation of vector embedding to learn transaction embeddings. Trans2Vec is fully unsupervised and learns a transaction embedding from both items and FIs. In this paper, we present Hui2Vec, a new approach that extends Trans2Vec to learn transaction embedding from items and HUIs.

3 Framework

3.1 Problem Definition

We follow the same notations as in [2] and in [5]. Given a set of items $I = \{i_1, i_2, ..., i_M\}$, a transaction dataset $D = \{T_1, T_2, ..., T_N\}$ is a set of transactions where each transaction T_i is a set of distinct items (i.e., $T_i \subseteq I$).

Our goal is to learn a mapping function $f : D \rightarrow \mathbb{R}^d$ such that every transaction $T_i \in D$ is mapped to a d-dimensional continuous vector. The mapping needs to capture the similarity among the transactions in D, in the sense that T_i and T_j are similar if $f(T_i)$ and $f(T_j)$ are close to each other in the vector space, and vice versa. The matrix $\mathrm{X} = [f(T_1), f(T_2), ..., f(T_N)]$ then contains feature vectors of transactions, which can be direct inputs for many traditional machine learning and data mining tasks, particularly classification.

3.2 Learning Transaction Embeddings Based on Items

We follow the same methodology as in Trans2Vec [5] with a part of HUIM embedding coming from singleton items directly. We adapt the Vector-Distributed Bag-of-Words (PV-DBOW) [15] where each transaction is a document and items are the words. Given a transaction T_t, and a set of items $I(T_t) = \{i_1, i_2, ..., i_k\}$

contained in T_t, we need to learn the representation of T_t by maximizing the log probability of predicting i_1, i_2, ..., i_k appearing in T_t :

$$max\Sigma_{j=1}^{k} logP_r(i_j \mid T_t) \tag{1}$$

$\Pr(i_j \mid T_t)$ is defined by a softmax function:

$$P_r(i_j \mid T_t) = \frac{exp(g(i_j).f(T_t))}{\Sigma_{i' \in I} exp(g(i').f(T_t))} \tag{2}$$

where $g(i_j) \in R^d$ is the embedding vector of item i_j and $f(T_t) \in R^d$ is the embedding vector of the transaction T_t.

We follow the negative sampling technique in Word2Vec [14] to avoid iterating on all items in I and instead select a small number of items not contained in the transaction T_t. The problem become in the form of minimizing this binary objective function of logistic regression:

$$\mathcal{O}_1 = -[log\sigma(g(i_j).f(T_t)) + \sum_{n=1}^{k} \mathbb{E}_{i^n \sim P(i)} log\sigma(-g(i^n).f(T_t))] \tag{3}$$

Where $\sigma(x) = \frac{1}{1+e^{-x}}$ is a sigmoid function, $P(i)$ is the negative item collection, i^n is a negative item taken from $P(i)$ for K times, and $g(i^n) \in \mathbb{R}^d$ is the embedding vector of i^n.

The derived gradients resulting from minimizing the objective function in Eq. 3 using stochastic gradient descent (SGD) are the following:

$$\frac{\partial \mathcal{O}_1}{\partial g(i^n)} = -\sigma(g(i^n).f(T_t) - \mathbb{I}_{ij}[i^n]).f(T_t)$$

$$\frac{\partial \mathcal{O}_1}{\partial f(T^t)} = -\sum_{n=0}^{k} \sigma(g(i^n).f(T_t) - \mathbb{I}_{ij}[i^n]).g(i^n) \tag{4}$$

where $\mathbb{I}_{ij}[i^n]$ is a function indicating whether i^n is an item i_j (i.e., the negative item is contained in the transaction T_t), and if $n = 0$, then $i^n = i_j$.

3.3 Learning Transaction embedding Based on High Utility Itemsets

As mentioned in section 1, we take for hypothesis that using HUIs instead of FIs would be more effective since HUIs would capture more information and are generally more appropriate for a wide range of applications such as market basket analysis, clickstream analysis, and bio-medicine. With Hui2Vec, we learn transaction embeddings based on HUIs instead of FIs, as was the case with Trans2Vec. We believe that the transaction representation learned with Hui2Vec will be more meaningful and discriminative than with Trans2Vec, especially in applications where HUIs are more informative than FIs.

Table 1. A Sample Transaction Database

TID	Transaction
T_1	(a,1) (c,1) (d,1)
T_2	(a,2) (c,6) (e,2) (g,5)
T_3	(a,1) (b,2) (c,1) (d,6) (e,1) (f,5)
T_4	(b,4) (c,3) (d,3) (e,1)
T_5	(b,2) (c,2) (e,1) (g,2)

The adaptation we make here to learn embeddings based on HUIs is very similar to the work made in Trans2Vec to learn Transaction Embeddings Based on Frequent Itemsets [5]. We follow the same notations in [6] and define a high utility itemset as follows. Let I be a finite set of items (symbols). An itemset X is a finite set of items such that $X \subseteq I$. A transaction database is a multiset of transactions $D = \{T_1, T_2, ..., T_N\}$ such that for each transaction T_c, $T_c \subseteq I$ and T_c has a unique identifier c called its TID (Transaction ID). Each item $i \in I$ is associated with a positive number $p(i)$, called its external utility. The external utility of an item represents its relative importance to the user. Every item i appearing in a transaction T_c has a positive number $q(i, T_c)$, called its internal utility. In the context of market basket analysis, the external utility typically represents an item's unit profit (for example, the sale of 1 unit of bread yields a 1 dollar profit), while the internal utility represents the purchase quantity of an item in a transaction (for example, a customer has bought 2 units of bread).

Table 2. External utility values

Item	a	b	c	d	e	f	g
Profit	5	2	1	2	3	1	1

Example. Consider the example transaction dataset in Table 1 with five transactions (T_1,T_2...T_5). Transaction $T2$ indicates that items a, c, e and g appear in this transaction with an internal utility (e.g. purchase quantity) of respectively 2, 6, 2 and 5.

Table 2 indicates that the external utility (e.g. unit profit) of these items are respectively 5, 1, 3 and 1. The utility of an itemset X is denoted as $u(X)$ and defined as u(X,T_c) = $\sum_{i \in X} u(i, T_c)$ if X $\subseteq T_c$. Otherwise u(X, T_c) = 0. The utility of an itemset X is denoted $u(X)$ and defined as $u(X) = \sum_{T_c \in g(X)} u(i, T_c)$, where g(X) is the set of transactions containing X. It represents the profit generated by the sale of the itemset X. For example, the utility of the item a in the transaction T_2 is u(a, T_2) = 5 × 2 = 10. The utility of the itemset {a, c} in T_2 is u({a, c},T_2) = u(a,T_2) + u(c,T_2) = 5×2 + 1×6 = 16. The utility of the itemset {a, c} is u({a, c}) = u(a)+u(c) = u(a, T_0)+ u(a, T_2) + u(a, T_3) + u(c, T_0) +

u(c, T_2) + u(c, T_3) = 5 + 5 + 10 + 1 + 1 + 6 = 28.

Table 3. High Utility Itemset for $minutil = 30$

HUI	Itemset	Utility
$X1$	{b, d}	30
$X2$	{a, c, e}	31
$X3$	{b, c, d}	34
$X4$	{b, c, e}	31
$X5$	{b, d, e}	36
$X6$	{b, c, d, e}	40
$X7$	{a, b, c, d, e, f}	30

The problem of high-utility itemset mining is to discover all high-utility itemsets, given a threshold $minutil$ set by the user. If $minutil = 30$, the high-utility itemsets or HUIs in the database of the running example are shown in Table 3. Each Transaction now can now be represented by a set of HUIs, as shown in Table 4.

Table 4. Each Transaction represented by a set of HUIs

TID	HUIs
T_1	{ }
T_2	$\{X_2\}$
T_3	$\{X_1, X_2, X_3, X_4, X_5, X_6, X_7\}$
T_4	$\{X_1, X_3, X_4, X_5, X_6\}$
T_5	$\{X_4\}$

Following the same procedure in the previous section, given a set of HUIs $F(T_t) = \{X_1, X_2, ..., X_l\}$ contained in a transaction T_t based on its HUIs is defined as follows:

$$\mathcal{O}_2 = -[log\sigma(h(X_j).f(T_t)) + \sum_{n=1}^{k} \mathbb{E}_{X^n \sim P(X)} log\sigma(-h(X^n).f(T_t))] \quad (5)$$

where $h(X_j) \in \mathbb{R}^d$ is the embedding vector of the frequent itemset $X_j \in F(T_t)$, $P(X)$ is the negative item collection (i.e., a small set of random HUIs which are not contained in T_t), X^n is a negative item taken from $P(X)$ for K times, and $h(X^n) \in \mathbb{R}^d$ is the embedding vector of X^n. Here again the objective function in Eq. 5 is minimized using SGD.

3.4 Hui2Vec Methods to learn Transaction Embeddings

When representing each transaction by a set of HUIs, we can face a situation where a Transaction T_t does not contain any HUIs. This is the case for example with transaction T_1 in Table 4. In this case, learning a useful embedding is very difficult. To overcome this problem, a good idea could be to combine information of both items and HUIs to learn embedding vectors for transactions. Here again, we follow the same model implemented in Trans2Vec [5].

Individual-Training Model to learn Transaction Embeddings. The method followed here, as illustrated in Fig.1, learns an embedding vector $f_1(T_t)$ for the transaction T_t based on its items and an embedding vector $f_2(T_t)$ based on its HUIs. The average of the two embedding vectors is taken as final embedding vector for the transaction $f(T_t) = \frac{f_1(T_t)+f_2(T_t)}{2}$.

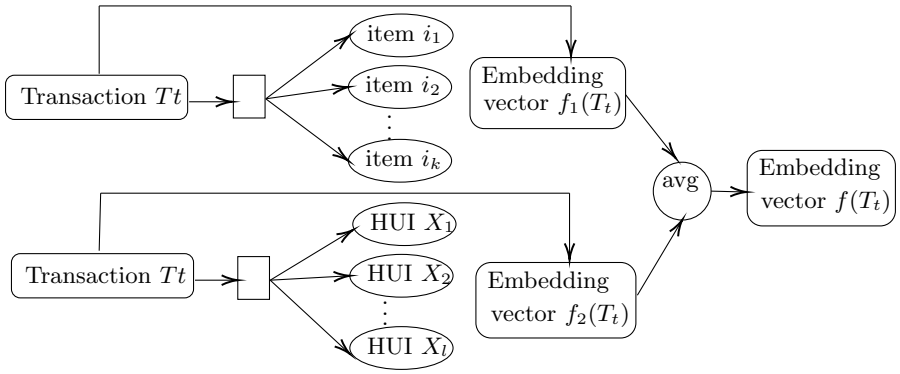


Fig. 1. The Individual Training Model of Hui2Vec

Joint-Training Model to Learn Transaction Embeddings. In the designed method, shown in Fig.2, both information from items and HUIs are combined. More precisely, given a Transaction T_t, the objective function that needs to be minimized is the following:

$$\mathcal{O} = -[\sum_{i_j \in I(T_t)} logPr(i_j \mid T_t) + \sum_{X_j \in F(T_t)} logPr(X_j \mid T_t)] \tag{6}$$

where $I(T_t)$ is the set of singleton items in T_t and $f(T_t)$ is the set high-utility items or HUIs contained in T_t. The equation 6 can be simplified to this form:

$$\mathcal{O} = -\sum_{p_j \in I(T_t) \cup F(T_t)} logPr(p_j \mid T_t), \tag{7}$$

where $p_j \subseteq T_t$ is an item or a high utility item. Usually, we refer to p_j as a pattern.

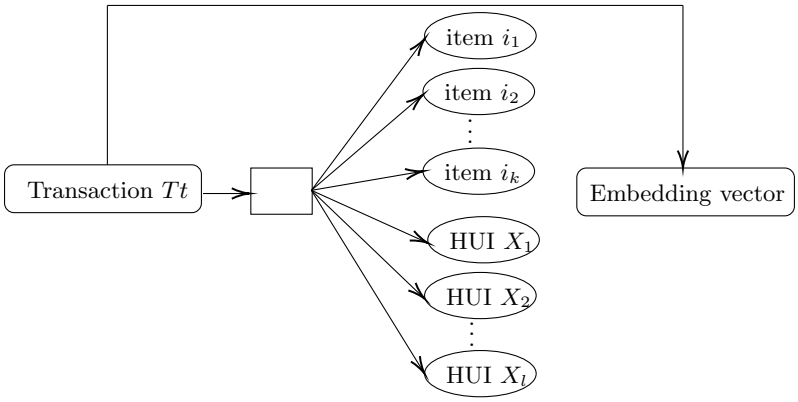


Fig. 2. The Joint-Training Model of Hui2Vec

The objective function here becomes the following:

$$\mathcal{O} = -[log\sigma q(p_j).f(T_t) + \sum_{n=1}^{k} \mathbb{E}_{p^n \sim P(p)} log\sigma(-q(p^n).f(T_t))] \tag{8}$$

where $q(p_j) \in \mathbb{R}^d$ is the embedding vector of the pattern $p_j \in I(T_t) \cup F(T_t)$, $P(p)$ is the negative pattern collection (i.e., random patterns which are not contained in T_t), p^n is a negative pattern drawn from $P(p)$ for K times, and $q(p^n) \in \mathbb{R}^d$ is the embedding vector of p^n. The minimization is here also made using SGD. After the learning process is completed, a transaction T_i will be close to a transaction T_j if they have similar items and HUIs.

4 Experiments

We conduct extensive experiments on real-world transaction datasets to evaluate Hui2Vec's performance for transaction classification.

4.1 Datasets

The statistics of the datasets used in the experiments are shown in Table 5. The columns 2 to 5 indicate the total number of transactions, the number of transactions for training and testing, the number of distinct items, the average transaction length, and the number of classes. All these datasets are taken from SPMF [17], an Open-Source Data Mining Library.

Foodmart_utility: dataset of customer transactions from a retail store, obtained and transformed from SQL-Server 2000.

Chainstore_utility: dataset of customer transactions from a major grocery store chain in California, USA, containing 1,112,949 transactions and 46,086 items, obtained and transformed from NU-Mine Bench.

ECommerce_retail_utility_no_timestamps: transactions recorded between 01/12/2010 and 09/12/2011 by a UK-based online retail store.

Liquor_11: This is a dataset of 9,284 customer transactions from US liquor stores from the state of IOWA. The dataset contains only transactions having no more than 11 items.

Table 5. Statistics of the datasets used in the experiments

Dataset	#transactions	#train	#test	#items	avg. length	#classes
Foodmart_utility	4,141	3,313	828	1,559	4.4	2
Chainstore	1,112,949	890,359	222,590	46,086	7.2	2
ECommerce_utility	14,975	11,980	2,995	3,468	11.71	2
Liquor_11	9,284	7,427	1,857	2,626	2.7	3

4.2 Implemented Models

As mentioned in the previous section, we implemented two different models for the proposed Hui2Vec method to learn transaction embeddings based on items and HUIs. The first model **Hui2Vec-IND** learns transaction embeddings from items and HUIs separately and then take the average. The second model **Hui2Vec-JOI** learns transaction embeddings from items and HUIs simultaneously. To find high utility itemsets in a dataset, we use EFIM's [6] implementation from the SPMF library [17].

To validate the efficiency of the designed Hui2Vec models, we compare them with the two corresponding models from the Trans2Vec approach [5], **Trans2Vec-IND** and **Trans2Vec-JOI**. Again, Trans2Vec models learn transaction embeddings from items and FIs. **Trans2Vec-IND** behaves in a way very similar to **Hui2Vec-IND** but uses HUIs instead of FIs. The same with **Trans2Vec-JOI** and **Hui2Vec-JOI**.

4.3 Evaluation Metrics

To evaluate the proposed models, we follow the same methodology as in [5]. Once embeddings of transactions are learned, we feed them to an SVM with linear kernel [31] to classify the transaction labels. We use the linear-kernel SVM (a simple classifier) and do not tune the parameter C of SVM (here, we fix $C = 1$) since our focus is on the transaction embedding learning, not on a classifier. Each dataset is randomly shuffled and split into the training and test sets as shown in Table 1. All methods are applied to the same training and test sets. We repeat the classification process on each dataset ten times and report the average classification accuracy and the average F1-macro score. We do not report the standard deviation since all methods are very stable (their standard deviations are less than 10^{-2}).

4.4 Parameter Settings

Our method Hui2Vec has two important parameters: the high-utility threshold *minutil* for extracting HUIs and the embedding dimension d. Trans2Vec the method we are comparing our model to has also two important parameters: the minimum support threshold δ for extracting FIs and the embedding dimension d. Since we are developing a fully unsupervised approach, all these values are assigned without using transaction labels. We set $d = 128$ which is a common value used in embedding methods [5]. We set δ and *minutil* following the *elbow method* [32, 5].

Table 6. Experiments results - Comparison between *Trans2Vec* and *Hui2Vec* in Accuracy (AC) and F1-macro (F1) on the four datasets.

	Foodmart_utility		Chainstore		Ecommerce_utility		Liquor_11	
Method	AC	F1	AC	F1	AC	F1	AC	F1
Trans2Vec IND	50.1	49.77	43.09	42.81	50.27	47.50	43.67	43.50
Trans2Vec JOI	67.06	66.89	52.90	52.74	82.52	82.36	81.15	81.45
Hui2Vec IND	92.9	92.55	80.62	79.55	94.76	94.42	93.34	93.57
Hui2Vec JOI	**93.58**	**93.33**	**81.43**	**80.46**	**95.43**	**95.11**	**94.25**	**94.83**

4.5 Results and Discussion

Results are shown in Fig. 6. Two main observations are made. First, we can see that the two models of the proposed Hui2Vec method are clearly providing better classification on all datasets compared to the two other models from Trans2Vec. **Hui2Vec-JOI** improved accuracy over **Trans2Vec-JOI** on all datasets by large margins (achieving 15-52 % gains). This demonstrates that the proposal of incorporating information of both singleton items and HUIs into the transaction embedding learning is a better strategy than learning transaction embeddings from combining items and FIs. This is reasonable because Trans2Vec relies on FIs and thus can only only capture frequent relationships between items, while Hui2Vec considers not only relationships between items but also the utility information (unit profit and quantities). This is interesting as utility information is generally available in real transaction datasets such as those used in the experiments. Hence, it is expected that in pratice, no additional effort is needed for users to collect data to apply Hui2Vec compared to Trans2Vec.

Second, we find that **Hui2Vec-JOI** produces better results than **Hui2Vec-IND** on all datasets. This confirms the idea that transaction embeddings learned from items and HUIs simultaneously are more meaningful and discriminative since they can capture different latent relationships of transactions simultaneously. Similarly, it is also observed that **Trans2Vec-JOI** always perform better than **Trans2Vec-IND** on all datasets.

On overall, experiments show good classification performance using the proposed Hui2Vec method to learn transaction embeddings.

5 Conclusion

In this paper, we demonstrated that learning transaction embeddings from both singleton items and HUIs produces more meaningful and discriminative representations than learning from items with FIs. We presented Hui2Vec, an unsupervised method for learning transaction embeddings from combining both singleton items and HUIs. We implemented two different models of the Hui2Vec approach. Experiments were carried out on four datasets, which showed that the Hui2Vec models significantly outperform the Trans2Vec models [5], which learns from items and FIs, in both accuracy and F1-macro scores.

As an extension to this work, we plan on studying the impact of parameters on the method (the high-utility threshold *minutil* for extracting HUIs and the embedding dimension d) on the quality of the embeddings produced. Another extension is to investigate the quality of the embeddings for the transaction clustering task.

Acknowledgment

Authors would like to thank the authors of Trans2Vec [5] for providing their source code. This work is partially supported by NARD Intelligence[1].

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